

Newton's method.

1. 原始牛顿法: 求方程的根: 如: $x^3 - 2x - 5 = 0$, $e^x + x = 0$, $\cos x = x$ $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
2. 最优化中的牛顿法: 求解无约束优化问题. $x_{n+1} = x_n - [\nabla^2 f(x_n)]^{-1} \nabla f(x_n)$

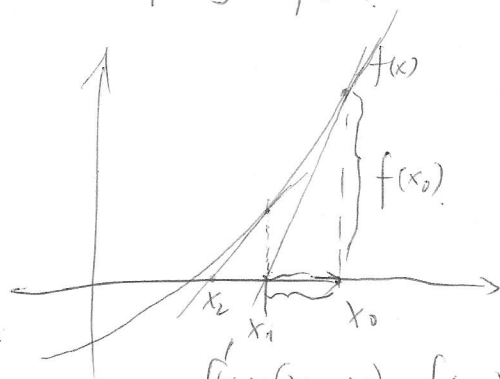
例子: $f(x) = x^2 - 2 = 0$

$$x_{n+1} = x_n - \frac{x_n^2 - 2}{2x_n} = \frac{x_n + \frac{2}{x_n}}{2}$$

$$x_0 = 1$$

$$x_1 = 1.5$$

$$x_2 = 1.41667 \dots$$



The goal of Newton's method for estimating a solution of an equation $f(x) = 0$ is to produce a sequence of approximations that approach a solution.

$$\begin{aligned} f(x_0) \cdot (x_0 - x_1) &= -f(x_0) \\ \Rightarrow x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} \end{aligned}$$